MIL-T-58078A(AV)
25 February 1985
Superseding
MIL-T-58078(MO)
16 July 1964

MILITARY SPECIFICATION

Tester, Pitot and Static Systems

This specification is approved for use by US Army Aviation Systems Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers an electric motor-driven, wet pump type, vacuum and pressure test set.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

MH-T-791 - Tube, Flexible, Nonmetallic, Electrical.
QQ-P-416 - Plating, Cadmium (Electrodeposited).
TT-E-489 - Enamel, Alkyd, Gloss (for Exterior and Interior Surfaces).
WW-T-700/3 - Tube, Aluminum Alloy, Drawn, Seamless, 2024.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple, Wall.

Military

MIL-P-116 - Preservation, Methods of.
MIL-F-5509 - Fittings, Flared Tube, Fluid Connection.
MIL-H-5593 - Hose, Aircraft, Low-Pressure, Flexible.
MIL-S-6872 - Soldering process, General Specification for.

Beneficial comments (Recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Aviation Systems Command, ATTN: AMSAV-ELSS, St. Louis, MO 63120-1798, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

NO DELIVERABLE DATA REQUIRED BY THIS DOCUMENT

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MIL-C-6985 - Clamps, Hose.
MIL-I-7721 - Indicator, Rate of Climb.
MIL-M-8609 - Motors, Direct-Current, 28-Volt System, Aircraft Class A and B, General Specification for.
MIL-A-8625 - Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-C-13924 - Coating, Oxide, Black, for Ferrous Metals.
MIL-A-27229 - Altimeter, Pressure AAU-8/A.
MIL-T-28800 - Test Equipment for use with Electrical and Electronic Equipment, General Specification For

STANDARDS

Federal

FED-STD-595 - Colors.

Military

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-143 - Specifications and Standards Order of Precedence for the Selection of.
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-810 - Environmental Test Methods for Aerospace and Ground Equipment.
DOD-STD-1866 - Soldering Process General (Non-Electrical)
MS3102 - Connector, Receptacle, Electric, Box Mounting.
MS24123 - Plate, Identification.
MS27404 - Adapter, Straight, Reusable, Tube to Hose, Low Pressure.
MS28046 - Indicator, Indicated Airspeed, Pitot Static, 40-400 Knots.
MS33558 - Numerals and Letters, Aircraft Instrument Dial, Standard Form of.
MS33584 - Tubing End - Standard Dimensions for Flared.
MS33586 - Metals, Definition of Dissimilar.
MS33611 - Tube Bend Radii.
MS33620 - Chart, Hose Selection.

Air Force-Navy Aeronautical

AN6270 - Hose Assembly - Detachable Swivel Fitting, Low Pressure.

Air Force-Navy Aeronautical Design Standard

AND10059 - Fittings - Styles of Pipe Threaded, Flared Tube and Hose.

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DRAWINGS

US Army Aviation Research and Development Command

SK-ED-0017 - Flush Static Port Adapter.
SK-ED-0018 - Pitot Head Adapter.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procurement activity or as directed by the contracting officer).

2.2 Other Publications. --The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society of Testing and Materials
ASTM B633 Zinc On Iron & Steel, Electrodeposited Coatings Of

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19105).

National Bureau of Standards

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the United States Department of Commerce, National Bureau of Standards, Washington, 25 DC).

National Fire Protection Association National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269).

3. REQUIREMENTS

3.1 Tester. The pitot and static systems tester shall be a single integral unit consisting of a carrying case with accessories; motor pump section for supplying vacuum and pressure; pneumatic switching valves; master pneumatic indicating gages and safety devices for protecting the gages; and containers suitable for storing high-temperature oils.

3.1.1 Preproduction Sample. One tester shall be subjected to the preproduction tests specified in 4.2.1. A preproduction sample is defined as a production line sample of the product which the manufacturer will subsequently produce in fulfillment of the contract. Approval of the preproduction sample shall be obtained from the procuring activity prior to production of items in fulfillment of the contract or order. Approval of the preproduction sample by the procuring activity in no way relieves the contractor of his responsibility

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to produce testers which meet all requirements of this specification.

3.2 Design. Tester shall be designed to test the complete pitot and static systems and instruments on aircraft specified by the procuring activity (see 6.4). It shall be capable of simulating pressure and vacuum, equivalent to the readings listed in Table I, simultaneously, to the instrument being tested and the counterpart master instrument on the tester panel. The panel shall consist of the following:

Instrument	Range	Referenced Specification or Standard
Rate of Climb	0 to ± 6,000 feet per minute	MIL-I-7721
Altimeter pressure	0 to 35,000 feet	MIL-A-27229
Airspeed indicator	40 to 400 knots	MS28046

General appearance and configuration of the tester shall be in accordance with Figure 1. Instrument panel and valve arrangement shall be as shown in Figure 2. Instrument calibration cards similar to those shown on Figure 4 and a conversion chart showing knots per hour to miles per hour shall be mounted on the tester. The control valve access panel shall be hinged to the rear edge of the top of the tester. The instrument panel shall be hinged to the forward edge of the top of the tester in accordance with Figure 3. The instrument panel shall swing upward at least 2° above the horizontal plane formed by the top of the tester to provide convenient access to internal components. The panels shall be fitted with quick release fasteners. All panels include pertinent nomenclature and warning notes to prevent improper use of the controls. All selector valves, external hose connections, and the bellows assembly block shall be labled with the appropriate nomenclature.

3.2.1 Design Capacities.--The tester shall be capable of evacuating a total system volume comprised of 2,850 cc plus the tester's own volume. The tester shall be capable of evacuating the total system to produce the rates and/or altitudes necessary to perform the following tests.

a. Rate of Climb Pump Capacity.--The tester shall maintain a rate of climb up to 5,000 feet/minute for a period of three minutes.

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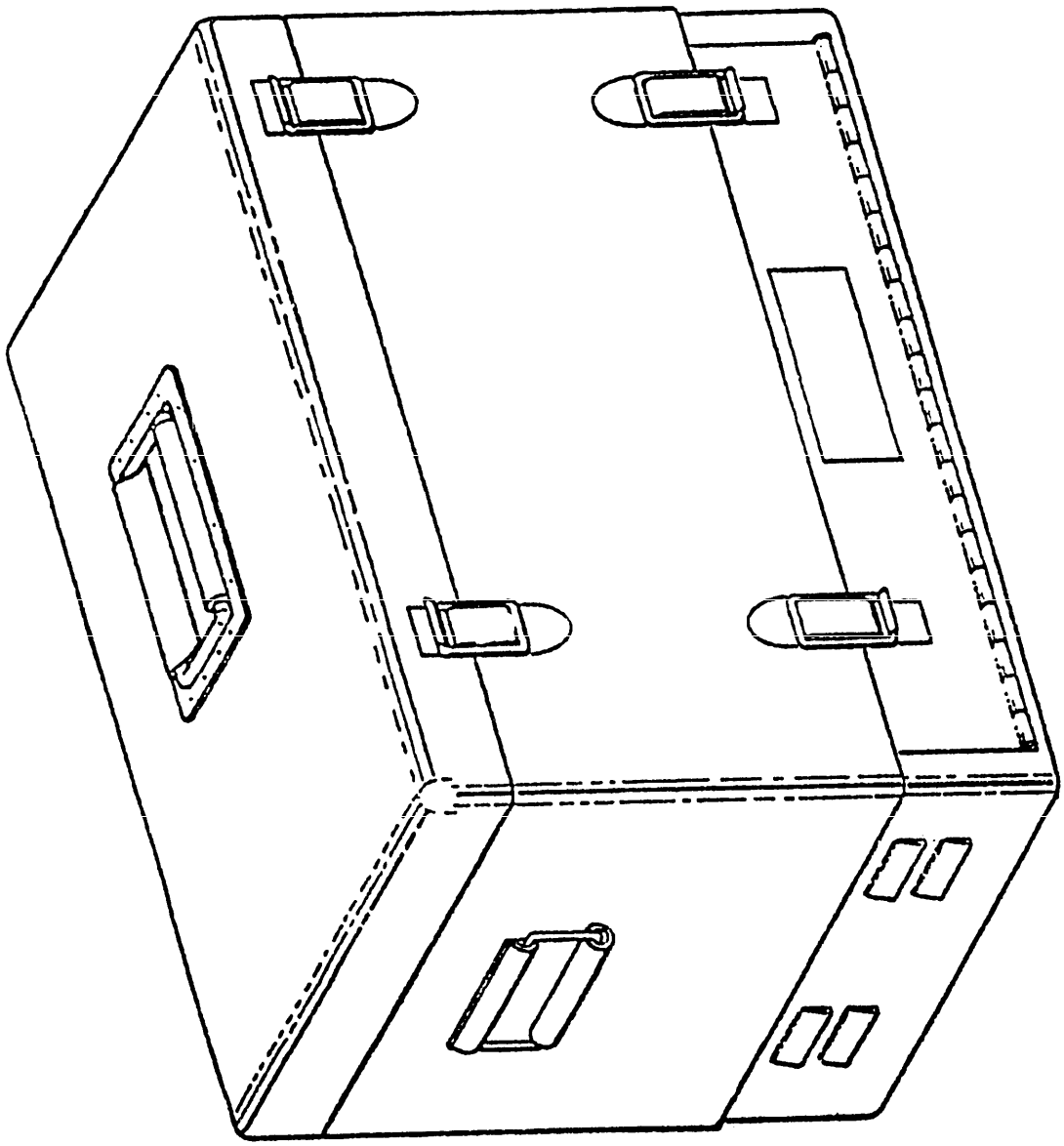


FIGURE 1. Overall view of tester, pitot and static systems in carrying case.

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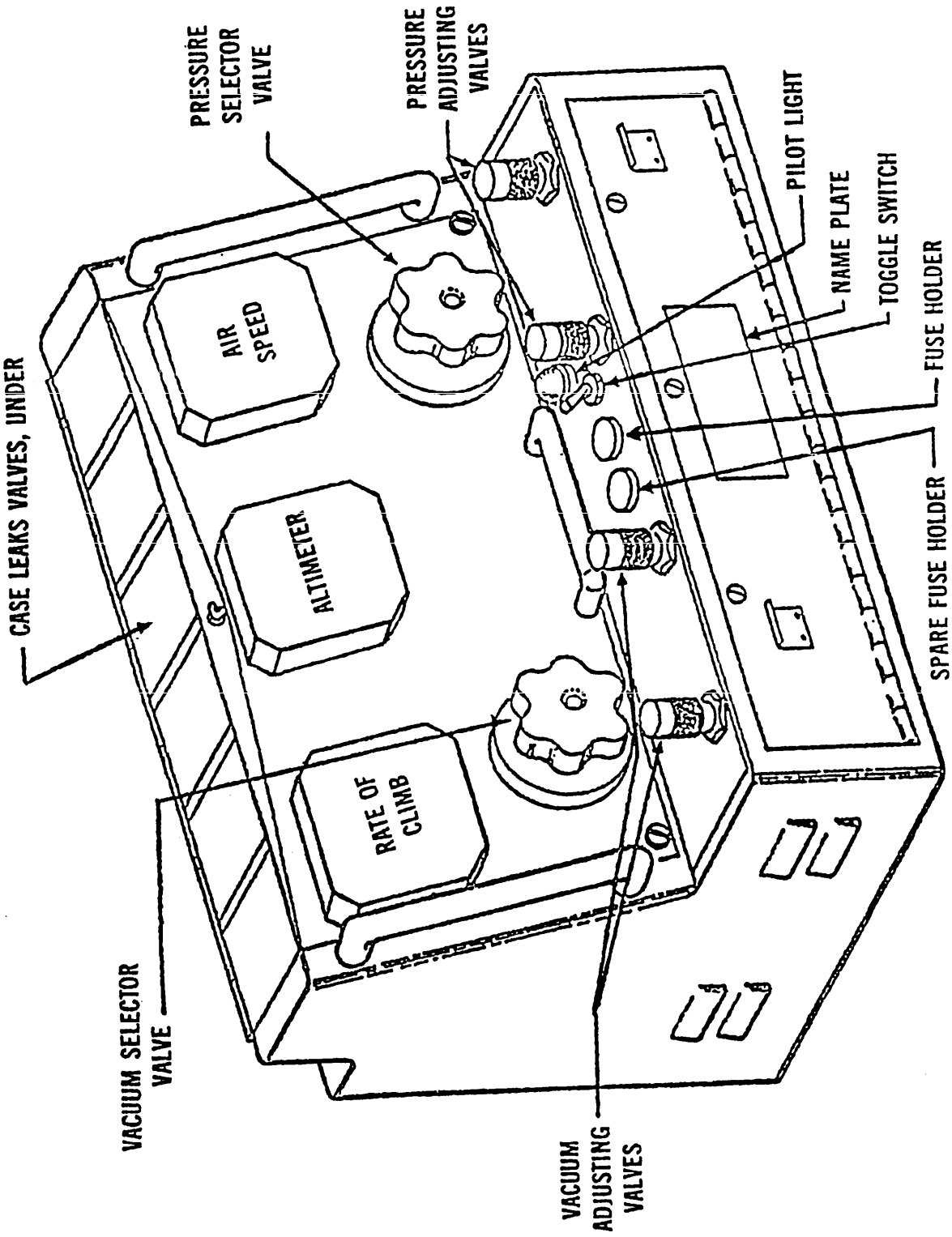


FIGURE 2. Control panel for tester, pitot and static systems.

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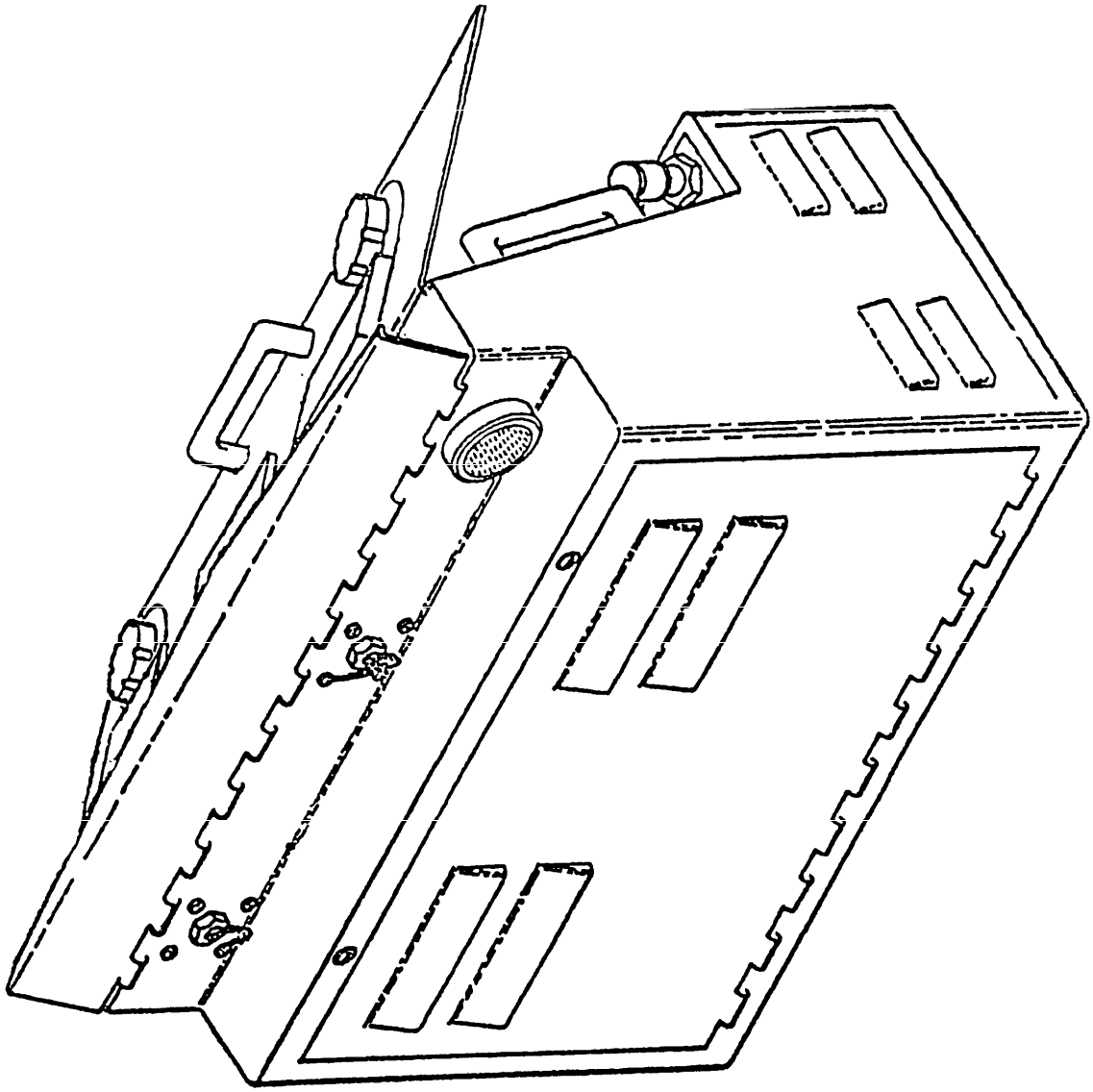


FIGURE 3. Hinged panel for tester, pitot and static systems.

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AIR SPEED INDICATOR			
STANDARD AIR SPEED IN KNOTS	INDICATOR READING IN KNOTS	STANDARD AIR SPEED IN KNOTS	INDICATOR READING IN KNOTS
40		220	
60		240	
80		260	
100		280	
120		300	
140		320	
160		340	
180		360	
200		400	
INST NO.		DATE	

ALTIMETER			
STANDARD ALTITUDE IN FEET	INDICATOR READING IN FEET	STANDARD ALTITUDE IN FEET	INDICATOR READING IN FEET
0		20000	
500		22000	
1000		25000	
2000		30000	
4000		35000	
6000			
8000			
10000			
12000			
14000			
16000			
18000			
INSTR. NO.		DATE	

RATE OF CLIMB INDICATOR				
CALIBRATION IS AT RATES OF CLIMB AT ALTITUDE INTERVALS SHOWN				
TEST RATE IN FEET PER MINUTE	RATE OF CLIMB		RATE OF DIVE	
	STOP WATCH ELAPSED TIME	INDICATED RATE	STOP WATCH ELAPSED TIME	INDICATED RATE
500 FT.				
1000 FT.				
2000 FT.				
3000 FT.				
4000 FT.				
5000 FT.				
6000 FT.				
INSTR. NO.			DATE	

FIGURE 4. Instrument calibration cards.

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b. Altimeter Pump Capacity.--The tester shall be capable of producing pressures equivalent to altitudes of -1,000 feet to 40,000 feet at the following rates and conditions:

Altitude (feet)	Rate of change in altitude (feet/minute)
0	500
1,000	500
3,000	500
5,000	500
10,000	500
20,000	3,000
30,000	3,000
35,000	3,000
40,000	3,000

3.3 Materials.--Materials shall be as specified herein. When specifications and standards are not specifically designated, selection of materials and processes shall be in accordance with Standard MIL-STD-143.

3.3.1 Subsitution of Materials.--Materials not conforming to the specifications and drawings referenced herein shall not be substituted unless such substitutions are specifically approved by the contracting officer. Substitutions will be approved only if it is clearly demonstrated that an improvement in operating characteristics, a saving in weight, conservation of critical or strategic materials, or a reduction in cost without sacrifice of reliability can be accomplished thereby, and that such substitutions will not preclude the subsequent use approval of such substitutions shall be accompanied by completely detailed supporting data which clearly illustrate the use of the proposed substitute materials.

3.4.1 Standard Parts--AN OR MIL standard parts shall be used whenever they are suitable for the purpose, and shall be identified on the drawings by their part numbers. Commercial utility parts such as screws, bolts, nuts, and cotter pins, etc., may be used provided they have suitable properties and are replaceable by AN or MIL standard parts without alteration. In applications for which no suitable correspondence AN or MIL part is in effect on date of invitation for bids, commercial parts may be used provided they conform to this specification.

3.4.2 Threaded Parts.--Threaded parts shall conform to Handbook H28. The American National fine thread series shall be used for threaded parts less than one-quarter inch diameter. The Unified or American National coarse thread series shall be used for threaded parts one-quarter inch diameter and larger; however, the American National fine thread series may be employed for these sizes when applicable. Pipe threads shall conform to the American National taper pipe thread series.

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3.4.3 Dissimilar Metals.--Unless components are suitably protected against electrolytic corrosion, contact between dissimilar metals shall be avoided. Dissimilar metals are defined in Standard MS33586.

3.4.4 Interchangeability.--Components shall be interchangeable (without requiring modification for replacement) with similar components furnished under the same contract or order.

3.4.5 Soldering.--Soldering shall be in accordance with Specification MIL-S-6872.

3.4.6 Adjustments.--All parts subject to adjustment or replacement by operating or maintenance personnel shall be readily accessible. Adjusting facilities, where provided, shall operate smoothly and without interference and shall adequately maintain such adjustments under all conditions of service. Designs which render servicing, adjustment, or replacement unduly difficult under field conditions will not be accepted. Valves, switches, and external connections shall be designed to facilitate operation when gloves are worn.

3.5 Performance.--Tester shall perform satisfactorily when subjected to tests specified in 4.2 and 4.3.

3.6 Components.

3.6.1 Carrying Case.--Carrying case to cover the tester shall be a drip-proof type as defined in Standard MIL-STD-108. Its design shall enable the tester to be removed from the case without the use of special tools.

3.6.1.1 Accessory Storage.--The case shall incorporate a compartment for storing accessory equipment in a manner to withstand the shock and vibration of handling and transporting the tester. The accessory storage compartment shall have a nondetachable lid and shall be accessible without removal of the tester from the case.

3.6.2 Handles.--Handles shall be provided to carry the tester with or without the case installed. Location and construction of handles shall be as shown on Figures 1 and 2.

3.6.3 Latches.--Latches shall secure the case to the tester, and the lid to the accessory compartment. The latches shall be protected against damage from contact with other objects. The latches shall support the weight of the equipment. The latches shall be designed to eliminate accidental opening during handling and transport of the equipment.

3.6.4 Feet.--Feet, or projections, shall be suitable to raise the under surface of the frame to prevent scratching during handling.

3.7 Accessories.

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3.7.1 Power Cables.--Power cables, extensions, and adapters shall conveniently connect the tester to any of the power sources listed in 3.8.2. The cables shall be designed to connect each power source to the corresponding electrical system in the tester. Power cables shall possess adequate grounding.

3.7.2 Hose and Connections.--Hose, clamps, couplings, sleeves, and reducers, as applicable, shall connect the tester to points in the aircraft plumbing, to the pitot pressure port located in the aircraft pitot head, and to the static pressure port which may protrude, or be flush located, on the outer surface of the aircraft. Adapters shall also be provided to connect the tester directly to the instrument cases. For bench-testing uninstalled instruments, valves shall be provided with the accessory hose connections to isolate the instruments being tested for leakage. Acceptable hose and fittings with their applicable specifications, standards, or drawings, are listed in Table II. Adapters shall be as shown in Figure 5, and dimensions shall be in accordance with Drawings SK-ED-0017 and SK-ED-0018.

3.8 Electrical Equipment.--All electrical wiring and equipment shall be installed in conformance with the National Electrical Code. Wiring shall be cabled, laced, or placed in insulating tubing conforming to Specification HH-T-791, and supported. Adequate grounding terminals shall be securely attached to the tester metal enclosure.

3.8.1 Connectors.--Electrical input connector shall conform to Standard MS3102 and shall be located at the rear of the tester.

3.8.1.1 Internal Connections.--Internal connections of the power cables and tester shall be such that when the appropriate cables (see 3.7.1) are used for the power sources specified, the motor is automatically connected for proper operations.

3.8.2 Power Sources.--The performance specified in 3.5 shall be obtained when the equipment is operated from any one of the following sources:

a. 28-volt direct current (dc), +4 or -6 volts.

b. 115-volt, ± 10 percent, 50 to 500 cycles per second, single-phase.

c. 115volt, ± 10 percent, 50 to 500 cycles per second, three-phase.

The operating power shall not exceed 120 watts. The tester shall contain a step-down transformer, rectifier, filter condenser, and electrolytic capacitor to facilitate operating from voltages specified above.

3.8.3 Overload Protection.--The electrical system shall be protected with fuses in accordance with Specification MIL-T-28800.

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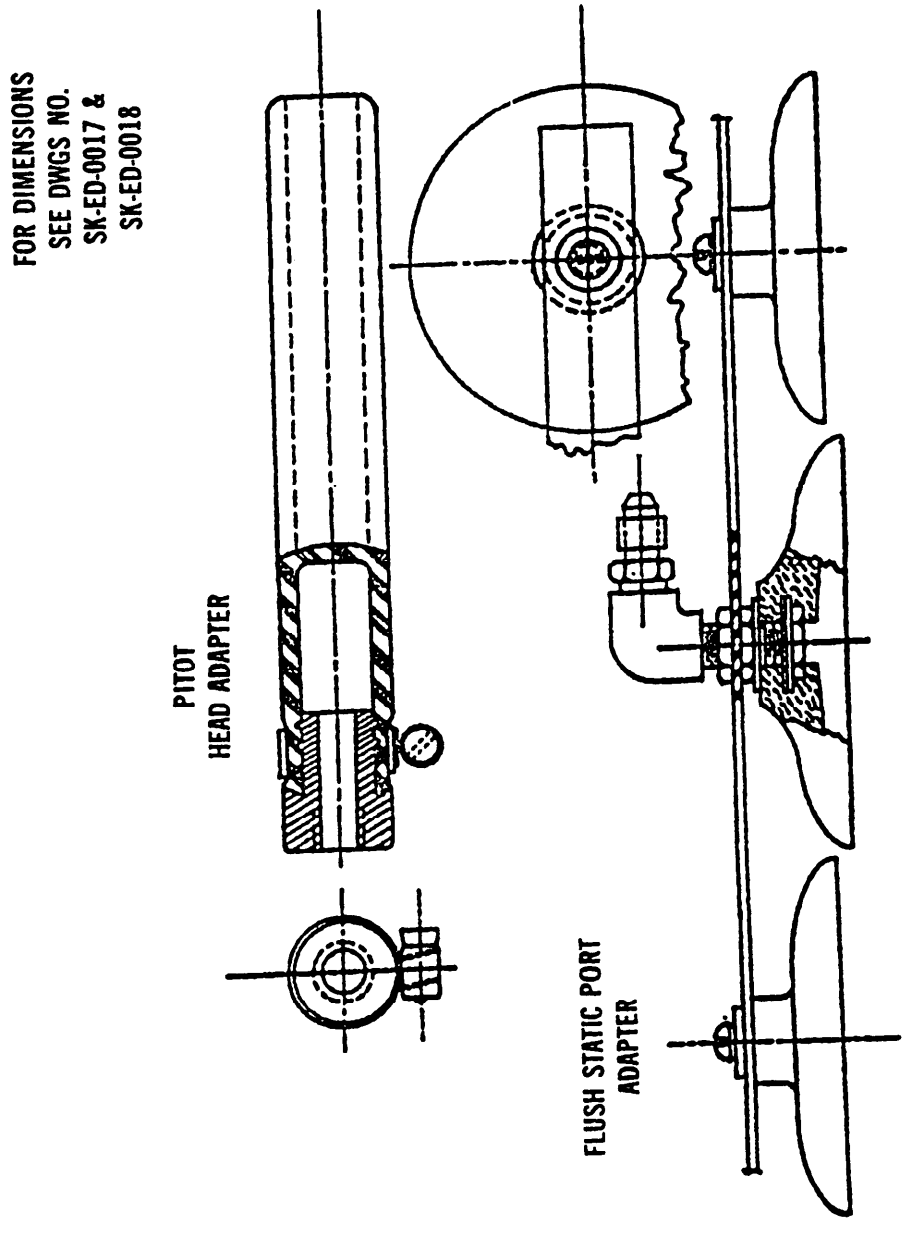


FIGURE 5. Adapters.

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3.8.4 Switch.--Application of power to the motor shall be controlled by an on-off switch located on the front panel. An indicator lamp shall indicate application of power to the motor.

3.9 Vacuum and Pressure Systems.--The vacuum and pressure systems shall consist of the following principal components:

3.9.1 Dual-Purpose Motor and Pump Assembly.--The motor section of the pump assembly shall be a conventional, high-speed, fractional horsepower, series wound motor conforming to Class A of Specification MIL-M-8609. The pump section shall be a direct drive, positive displacement, wet, rotary-type pump with multiple porting.

3.9.2 Air-Oil Reservoir.--An air-oil reservoir shall be connected to the pressure side of the pump. It shall separate the air and oil so that oil-free air is delivered to the instrument lines. The reservoir shall incorporate a transparent section or window, to facilitate visual inspection of oil level and condition.

3.9.3 Check Valves.--A check valve shall be located in both the vacuum and the pressure systems to prevent oil from entering the instrument lines during handling and transportation of the tester.

3.9.4 Oil Sump and Filter Assemblies.--Oil sump and filter assemblies shall be incorporated in the vacuum and pressure systems of the tester. The filters shall be of a design capable of removing any oil or contaminants that may enter the instrument lines. The sumps shall be transparent, to facilitate visual inspection for presence of oil.

3.9.5 Valves.

3.9.5.1 Adjusting.--Tester shall include a system of valves designed for adjusting the levels of vacuum or pressure with sufficient precision to maintain the readings and limits specified in Tables I and III.

3.9.5.2 Selector.--Two selector valves, one for the vacuum system and one for the pressure system, shall be provided. The selector valves shall consist of a central control manifold for setting up the desired test circuits. For each selector valve test position, all inlet and outlet pressure and vacuum lines, control valves, and instruments shall be properly connected.

3.9.5.3 Relief.--The tester shall incorporate relief valves to protect the instruments being tested and the instruments in the tester from excessive pressure or vacuum. The pressure and vacuum at which the relief valves operate shall be adjustable to the safety relief points specified in Table I.

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3.9.5.4 Case Leak.--Valves shall be incorporated as part of the tester to isolate each of the master instruments listed in 3.2 for calibration and "case leak" tests (see 4.3.3.4) without removal of instruments from the tester.

3.9.6 Vacuum and Pressure Lines.--Flexible pressure lines shall be used to allow the instrument panel to be raised to the desired position for access to components, and rigid pressure lines shall be used where flexibility is not required.

3.9.6.1 Flexible Pressure Lines.--Flexible pressure lines shall be hose assemblies consisting of materials and components and conforming to the following documents, as applicable.

a.	Hose assembly	AN6270 or MS33620
b.	Hose	MIL-H-5593
c.	Fittings assembly	MS27404
d.	Fittings	AND10059
e.	Fittings	MIL-F-5509

3.9.6.2 Rigid Pressure Lines.--Rigid pressure lines shall be assemblies consisting of tubing conforming to Specification WW-T-700/3 and fittings conforming to Specification MIL-F-5509. Tubing ends shall conform to Standard MS33584. Tubing bends shall conform to Standard MS33611.

3.10 Radio-Interference Suppression.--All radio-interference producing equipment shall be suppressed in accordance with Class B of MIL-STD-461.

3.11 Weight and Dimensions.

3.11.1 Weights.--Weight of tester, with carrying case and accessories, shall not exceed 60 pounds; without carrying case and accessories, shall not exceed 40 pounds.

3.11.2 Dimensions.--Overall dimensions, including carrying case, shall not exceed 20 by 18 by 18 inches; without carrying case, dimensions shall not exceed 19 by 17 by 17 inches.

3.12 Finish.--External surfaces shall be finished with gloss enamel conforming to Class B of TT-E-489. Color shall be orange-yellow conforming to No. 13538 of Standard FED-STD-595. If parts require coating, coatings shall conform to the following:

Materiel	Specification
Cadmium	QQ-P-416
Zinc plating	ASTM B633
Anodizing of aluminum	MIL-A-8625
Oxide coating	MIL-C-13924

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3.13 Marking and Labeling.

3.13.1 Identification Plate.--Tester shall be identified by a nameplate in conformance with Standard MS24123.

3.13.2 Marking.--All instrument marking shall be durable to withstand wear encountered in service. Letters and numerals shall conform to Standard MS33558. Numerals shall distinctly indicate the graduation to which each applies.

3.14 Drawings and Instruction Books.--Drawings and instruction books shall be furnished as specified in the contract or order (see 6.2).

3.15 Workmanship.--The tester, including all components, parts, and accessories, shall be fabricated and finished to produce an item free from defects which could affect proper functioning in service. Particular attention shall be given to neatness and thoroughness of soldering, welding, finish, wiring, marking, alignment of parts, tightness of assembly screws, bolts, rivets, and electrical connections and freedom of parts from blemishes, burrs, sharp edges, metal chips, loose solder, and foreign material.

TABLE I

Operating Tests

Instrument Indicator	Check Point	Safety Relief Point
Rate of climb	+5,000 ft fpm -5,000 ft fpm	
35,000 ft altimeter	-1,000 ft 25,000 ft 35,000	35,000 + 5,000 ft - 0000
Indicator airspeed	100 knots 200 knots 300 knots 400 knots	400 + 20 knots -0

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TABLE II

External Hose and Connections

Quantity	Nomenclature	Specification Standard or Drawing
1	Hose assembly 5 ft. (Lg)	AN6270-4
2	Hose assembly 6 ft. (Lg)	AN6702-4
A/R*	Reducer, 1/4 in to 3/16 in. external thread flared tube	MIL-F-5509
A/R*	Union, 1/4 in. flared tube	MIL-F-5509
A/R*	Clamp, sleeve	MIL-C-6985
1	Pitot head adapter	DWG SK-ED-0018
1	Flush static port adapter	DWG SK-ED-0017

*As required

TABLE III

System Leak Test

Instrument	Check Point Reading	Maximum Allowable System Leak Rate
Altimeter (1,000 ft.)	-1,000 ft.	50 ft./min.
Altimeter (35,000 ft.)	25,000 ft.	250 ft./min.
Indicator airspeed	200 knots	6 knots/min.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.--Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to

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perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of Examination and Tests.--Examinations and tests shall be classified as (1) preproduction and (2) conformance.

4.2.1 Preproduction.--Preproduction sample (see 3.1.1) shall be examined for conformance with the weights and dimensions specified in 3.11, for compliance with 4.2.2, and subjected to the tests specified in 4.3 through 4.3.6.

4.2.2 Conformance.--Each tester shall be examined and tested as follows:

4.2.2.1 Functional.

a. Operability of all hinges, covers, doors, and latches.

b. Compliance with 4.3 and 4.3.3.4.

4.2.2.2 Workmanship:

a. Presence of all parts and components.

b. Completeness and correctness of assemblies.

c. Presence and acceptability of identification plant and other required marking.

d. Fit at joints, seams, and edges.

e. Tightness of screws, bolts, rivets, and electrical connections.

f. Quality of finish.

4.3 Examination and Tests.

4.3.1 Conditions.

4.3.1.1 Atmospheric.--Tests shall be made at an atmospheric pressure of approximately 29.92 inches of mercury and at a temperature of 25° Centigrade (C) + 2° C. If tests are made with atmospheric pressure or temperature different from these values, allowances shall be made for the difference from the specified conditions.

4.3.1.2 Attitude.--The tester shall be tested in the normal operating position.

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4.3.2 Tapping.--The tester shall be gently tapped or vibrated while test readings are taken.

4.3.3 Procedures.

4.3.3.1 Performance.--The tester shall be operated on the three electrical power sources as specified in 3.8.2. During each phase of test the set shall be operated to produce the maximum positive and minimum negative pressures specified in Table I. The tester shall be operated to demonstrate conformance to the requirements of paragraph 3.2.1.

4.3.3.2 Operational.

4.3.3.2.1 Pressure (positive and negative).--With the test set operating, the applicable valve or control shall be operated to establish the check point and relief valve operating pressures specified in Table I.

4.3.3.3 System Leak.--Tester shall be operated to establish check point pressure specified in Table III. The proper valve shall be operated to retain this pressure in a static condition in the system. The operation of the tester shall be stopped and determination made of the system leakage, which shall not exceed that specified in Table III. This test shall be repeated as required to determine the leakage of each system.

4.3.3.4 Instrument Leak.--The tester shall be operated to obtain pressure specified as check point in Table III. The proper valve shall then be operated to isolate each instrument from the remaining system and hold this pressure in a static condition in the instrument. The leakage shall not exceed that allowed by the applicable specification or standard for instruments tested (see 3.2) Test shall be repeated as necessary to determine the leakage of each instrument.

4.3.3 Environmental.--The tester shall be placed within a test chamber maintained at a temperature of minus 40°C., plus or minus 2°C., for a 24-hour period. At the end of the 24-hour period and while still at minus 40°C., the tester shall be energized and meet the performance requirements of this specification (see 3.5).

4.3.4.2 High Temperature.--The tester shall be placed within a test chamber maintained at a temperature of 50°C., plus or minus 2°C., for a 24-hour period. At the end of the 24-hour period and while still at 50°C., the tester shall be energized and meet the performance requirements of this specification (see 3.5).

4.3.4.3 Storage Temperature.--The tester shall withstand, without damage, storage in temperature of minus 65°F., for a 72-hour period and 160°F., for a 72-hour period.

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4.3.4.4 Humidity.--The tester shall be subjected to the humidity test, method 507, procedure I of Standard MIL-STD-810. After completion of the test, the tester shall meet the performance requirements of this specification (see 3.5).

4.3.4.5 Salt Spray.--The tester shall be subjected to the salt spray test, method 509 of Standard MIL-STD-810. At the end of the test period the tester shall be examined for evidence of damage or detrimental effects.

4.3.5 Shock.--The tester shall be subjected to a shock test, method 516, procedure VI of Standard MIL-STD-810. After completion of the shock test, the tester shall meet the performance requirements of this specification (see 3.5).

4.3.6 Electromagnetic Interference Reduction.--Radio-interference reduction test shall be conducted as specified in MIL-STD-461 (see 3.10).

4.3.7 Materials.--It is not mandatory that all materials be tested in conformance with the referenced specifications in each individual case. However, the Government inspector will require these referenced tests whenever, in his judgement, they are necessary to ascertain that the quality of the materials used conforms to the referenced specifications.

4.4 Inspection of Preparation for Delivery.--Preservation, packaging, packing, and marking shall be inspected for conformance with section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging.--Preservation and packaging shall be level A or level C. as specified (see 6.2).

5.1.1 Level A.--Testers shall be individually preserved and packaged in accordance with method IIB of Specification MIL-P-116.

5.1.2 Level C.--Testers shall be preserved and packaged to prevent damage or deterioration during handling and shipment from the supplier to the first receiving activity.

5.2 Packing.--Packing shall be level A, level B, or level C, as specified (see 6.2).

5.2.1 Level A.--Testers preserved and packaged in accordance with 5.1 shall be individually packed in snug-fitting containers conforming to one of the following triple wall fiberboard box of class 2 of Specification PP-B-640; or cleate-plywood, oversea type box of Specification PPP-B-601; or nailed wood box of class 2 of Specification PPP-B-621. Items shall be cushioned to prevent movement within the shipping container. Box closure and strapping shall be in accordance with the applicable container specification or appendix thereto.

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5.2.2 Level B.--Unless otherwise specified, testers shall be packed as specified for level A (see 5.2.1), except that containers shall be domestic type or class, as applicable.

5.2.3 Level C.--Testers shall be packed in a manner to insure acceptance by common carrier for transportation at the lowest applicable rate and safe delivery to the first receiving activity.

5.3 Marking.--In addition to any special marking required by the contract or order (see 6.2), marking shall be in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended Use.--The tester is intended for use as a portable field instrument for checking the performance characteristics of vacuum and pressure aircraft instruments, pitot and static systems, and as a primary vacuum and pressure source for operation of aircraft computer systems.

6.2 Order Data.--Procurement documents should specify the following

a. Title, number, and date of this specification.

b. Whether drawings and instruction books should be furnished and if so, the specific requirements for each (see 3.14).

c. Level of preservation and packaging (see 5.1) and level of packing required (see 5.2).

d. Whether special marking is required (see 5.3).

e. A list of aircraft on which the tester will be use (see 6.4).

6.3 List of Aircraft.--The contracting officer should provide a current list of all aircraft on which this tester will be used.

6.4 Additional Features.--The requirements of this specification are a definition of the minimum design and performance characteristics which will be acceptable. Additional features which will increase performance, accuracy, or facilitate analysis or a large volume of samples may be submitted.

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a. Paragraph Number and Wording:

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